

Graph Representation

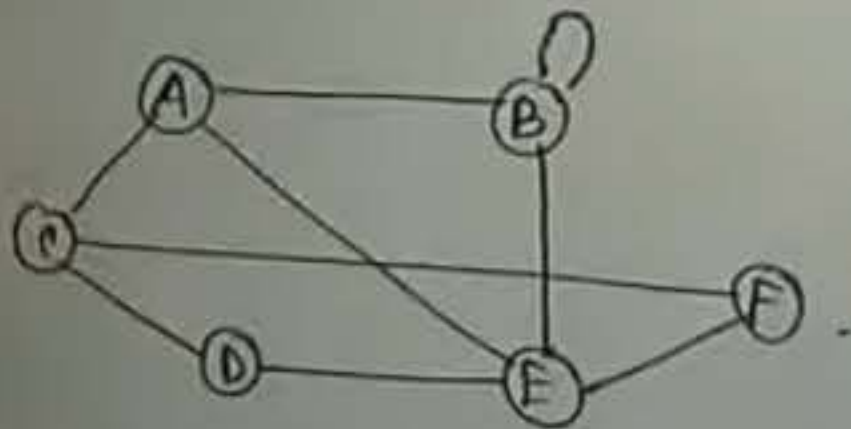
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- (i) Adjacency Matrix
- (ii) Incidence Matrix
- (iii) Adjacency List

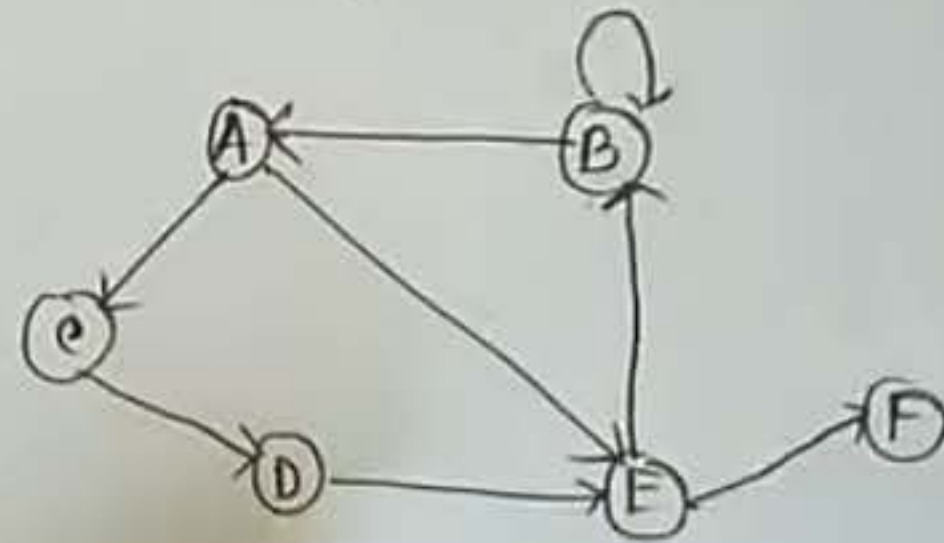
⇒ Undirected graph

- (i) no. of vertices (n) and edges (m)
- (ii) Square matrix represents graph
- (iii) $adj[i, j] = 1$ (loop, i, j adjacent are)
- (iv) $adj[i, j] = 0$ (i, j adjacent are not)

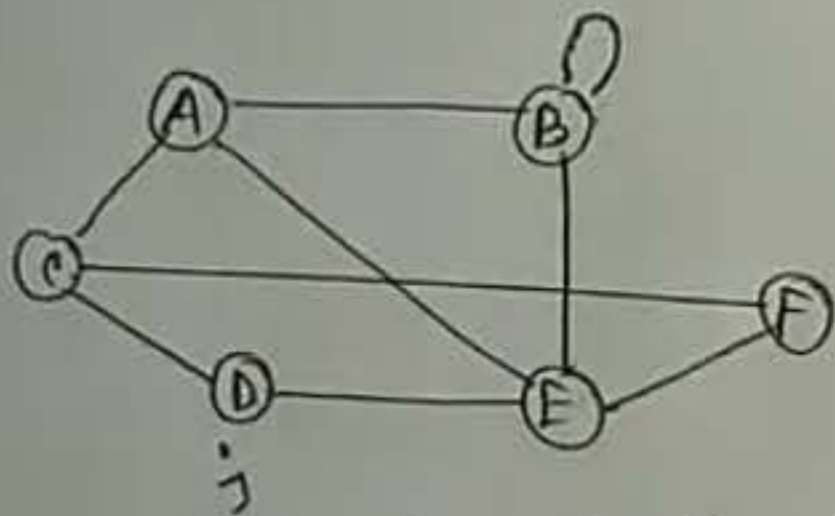


Adjacency Matrix

⇒ Directed graph



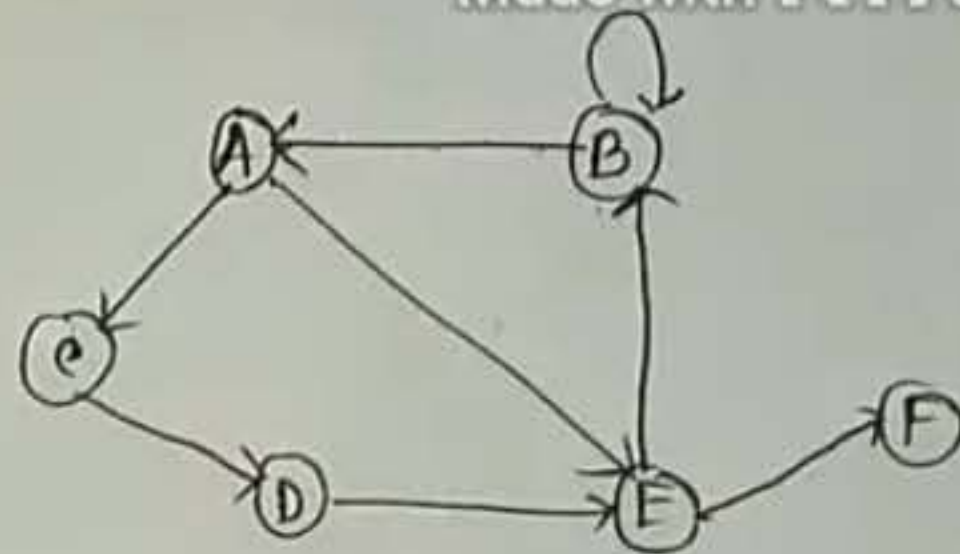
- ① no of vertices (n) $\Rightarrow$ $n \times n$ matrix
- ② Square matrix represents $n \times n$ matrix
- ③ $adj[i, j] = 1$ (loop i, j adjacent $\Rightarrow$ 1)
- ④ $adj[i, j] = 0$ (i, j adjacent $\Rightarrow$ 0)



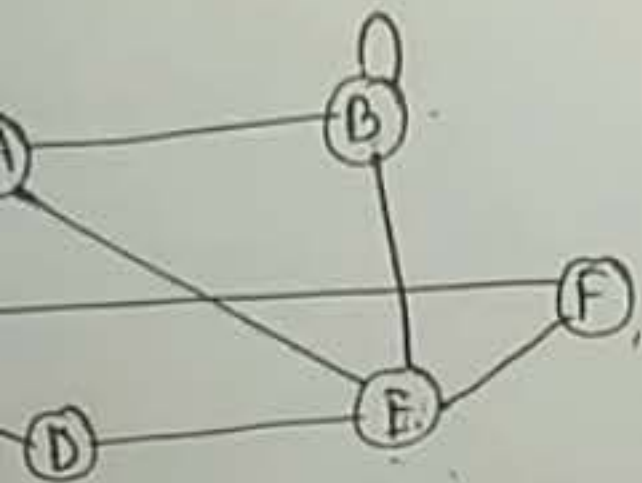
	A	B	C	D	E	F
A	0	1	1	0	1	0
B	1	1	0	0	1	0
C	1	0	0	1	0	1
D	0	0	1	0	1	0
E	1	1	0	1	0	1
F	0	0	1	0	1	0

$\Rightarrow$ Directed graph

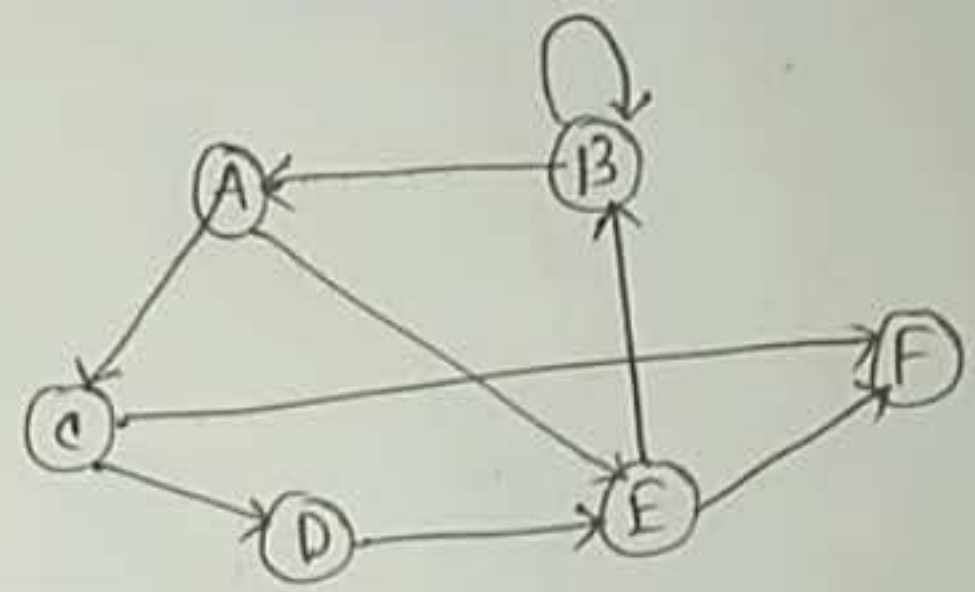
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vertices (m) $\times$ vertices (n) $\times$ edges (e)
 matrix represents graph $m \times m$
 1 (loop, i, i adjacent 2(H))
 0 (i, i adjacent not 2(H))



	A	B	C	D	E	F
A	0	1	1	0	1	0
B	0	1	0	0	1	0
C	1	0	0	1	0	1
D	1	0	0	1	0	1
E	0	0	1	0	1	0
F	0	0	1	0	1	0



	A	B	C	D	E	F
A	0	0	1	0	1	0
B	1	1	0	0	0	1
C	0	0	0	0	1	0
D	0	0	0	0	0	1
E	0	1	0	0	0	0
F	0	0	0	0	0	0